SENTRY-R: neural proposals and independent audits for privacy-risk release decisions

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Abstract

A release committee needs an explicit rule for deciding when the estimated success probability of a declared privacy attack is acceptable. A neural score alone cannot provide that rule. We propose SENTRY-R (scenario-enriched neural triage for release review), a workflow in which language models suggest attack scenarios, measured development attacks train a multi-output deep network, and an independent simultaneous audit decides release eligibility. For each candidate, risk is a vector indexed by adversary, harm, target group and release vintage, with a separate pooled-release attack. We derive the counted-outcome learning objective and backpropagation, then show that exact binomial upper bounds control the chance of any false certification across a finite shortlist even after adaptive development and utility-based final choice. On a synthetic toy law, over ten seeded replicates, a logistic model trained to convergence has slightly lower held-out log loss than a three-layer network in every replicate. The comparison shows that the software runs; it shows no gain from depth and no performance on real release programmes. The workflow exposes what a language model may change, what requires observed attack data, and which assumptions a release decision must record.

Keywords: statistical disclosure control; release governance; record linkage; neural risk model; language model; simultaneous inference.

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1. The release problem

Statistical agencies and central banks publish data because they have public value, but overlapping releases and auxiliary information may make records linkable. A reciprocal key frequency can quantify identification under a specific matching model [3]; it cannot silently stand in for every adversary, target group, time period or harm. Repeated access gives an attacker a different experiment from the one-release analysis. The underlying privacy-risk study examines this difference with a 200,000-record synthetic register, a granular credit-register simulation, two public microdata files and pooled-vintage attacks. Those empirical results precede the neural extension; none is a held-out evaluation of its deep model. The underlying derivations, experimental detail and caveats are provided in the accompanying full manuscript, *Governing privacy risk across repeated statistical releases: probability models, programme tolerances and accountable decisions*.

The added problem is methodological: how can an institution search a large family of release settings and adversaries while preserving a decision rule whose error rate survives that search? We specify a trainable proposal model and an independent audit. Utility remains an institutional value, whereas a policy tolerance is declared and signed by its owner. The price of harm and the value of the release can inform that tolerance [1]; a network cannot choose its legitimacy.

2. A risk vector with a declared sampling law

Let θ be a candidate release mechanism and $j = (a, h, g, v) \in \mathcal{J}$ encode an adversary, harm, target group and vintage. A pooled-vintage attack gets its own coordinate. The scenario specifies a binary success function $s_{\theta j}$ and a full law $P_{\theta j}$ for a fresh target, auxiliary information and attack randomness. Its risk and policy tolerance are

$$r_j(\theta) = \mathbb{E}_{P_{\theta j}} s_{\theta j}(Z), \qquad \tau_j \in (0, 1), \qquad |\mathcal{J}| = n. \tag{1}$$

A target-level probability and a group average are different estimands; both need their own scenario. After independent final audits yield simultaneous upper bounds $\overline{r}_{\theta j}$, define the audited margin vector and its minimum

$$z_j(\theta) = 1 - \frac{\overline{r}_{\theta j}}{\tau_j}, \quad \boldsymbol{z}(\theta) \in \mathbb{R}^n, \quad T(\theta) = \min_j z_j(\theta). \tag{2}$$

Release eligibility requires $T(\theta) \geq 0$ and the full vector must be reported. An unaudited declared coordinate receives upper bound one and fails when $\tau_j < 1$. The rule starts with no inherited claim of safety.

3. Scenario expansion and the deep proposal model

An LLM can suggest a new pooled attack or auxiliary information pattern. SENTRY-R treats its text and numeric features as *untrusted proposals*: a fixed schema checks names and ranges. The calling code records the model name and a hash of the exact prompt; the model's output cannot set these fields. A separate experiment observes successes K_{ij} in B_{ij} development attack trials. No generated success label can enter the training set. The validator and measured outcome are joined by a proposal digest. Human review remains responsible for the scenario's plausibility, lawful data access and sampling unit.

In a repeatable round, the analyst (i) freezes a feature and coordinate schema; (ii) accepts or rejects an LLM-suggested scenario with a recorded reason; (iii) runs the declared attack and writes independent development counts; (iv) appends measured rows to a new training version; and (v) evaluates on whole programmes omitted from training. Before any final audit, the software writes one hashed record that fixes the shortlist, α, the planned sample size of every audited coordinate, the tolerance owner, an expiry date and the identifiers of all development evidence. The audit reads α from that record and refuses a changed record, a lapsed expiry date, a trial count that departs from the plan, and evidence already used in development. An optional append-only ledger refuses a second audit of the same record. These checks bind an audit to its record. They do not prove what happened first: the hash has no key or trusted time stamp, and nothing stops an analyst from writing a new record or a new ledger. These steps permit many rounds of model improvement without making the subsequent confidence statement depend on a generated answer.

For $x_i \in \mathbb{R}^d$ and L ReLU hidden layers, the multi-output predictor is

$$h_i^0 = x_i, \quad h_i^{\ell+1} = \max\{0, h_i^\ell W_\ell + b_\ell\}, \quad \widehat{\boldsymbol{r}}_\phi(x_i) = \sigma(h_i^L W_{\mathrm{o}} + b_{\mathrm{o}}) \in (0, 1)^n. \tag{3}$$

Depth zero is logistic regression. Training minimises counted binomial negative log likelihood plus a declared ridge penalty:

$$\mathcal{L}(\phi) = -\frac{1}{\sum_{ij} B_{ij}} \sum_{ij} [K_{ij} \log p_{ij} + (B_{ij} - K_{ij}) \log(1 - p_{ij})] + \frac{\lambda}{2} \sum_\ell \|W_\ell\|_F^2. \tag{4}$$

If $K_{ij} \mid x_i, B_{ij} \sim \mathrm{Binom}(B_{ij}, \pi_j(x_i))$ and the trial count was chosen before observing successes, the conditional expected unregularised loss is minimised at $p_{ij} = \pi_j(x_i)$: its excess is a positive multiple of $D_{\mathrm{KL}}(\mathrm{Bern}(\pi_j) \| \mathrm{Bern}(p_{ij}))$. With $P = (p_{ij})$, the logit derivative is $(B \odot P - K)/\sum_{ij} B_{ij}$; multiplying backward through $W_\ell^\top$ and the ReLU indicator gives every hidden-layer gradient. This derivation states precisely what the software trains. Calibration on a held-out *programme* diagnoses prediction; it does not certify a new target [4].



Figure 1: The scenario proposal changes development work. A separate final attack audit decides release eligibility.

4. Independent audit and finite-family guarantee

The network ranks a fixed shortlist Θ_0 of C candidates using the largest predicted risk-to-tolerance ratio. For every (θ, j), freeze $B_{\theta j}$ and draw fresh, conditionally independent attacks; let $K_{\theta j}$ succeed. With $M = Cn$ and $\gamma = \alpha/M$, define the exact one-sided binomial upper bound [2]

$$\overline{r}_{\theta j} = \begin{cases} 1, & K_{\theta j} = B_{\theta j}, \\ F^{-1}_{\mathrm{Beta}(K_{\theta j}+1,\, B_{\theta j}-K_{\theta j})}(1-\gamma), & K_{\theta j} < B_{\theta j}. \end{cases} \tag{5}$$

The bound includes, rather than estimates away, final attack-draw uncertainty.

Proposition 1 (Selection after adaptive development)**.** *Let $\mathcal{D}$ contain all development information, including repeated LLM proposals, observed development attacks, fitted models and the shortlist. Conditional on $\mathcal{D}$, fix the release, scenarios, C, n and each sample size. If each final count has law $K_{\theta j} \mid \mathcal{D} \sim \mathrm{Binom}(B_{\theta j}, r_j(\theta))$, then selecting any candidate with $T(\theta) \geq 0$, including the most useful eligible candidate, has probability at most α of accepting any candidate whose actual declared risk exceeds a tolerance.*

Proof. For a fixed risk p and count $k < b$, the binomial lower tail $F_p(k) = \sum_{t=0}^{k} \binom{b}{t} p^t (1-p)^{b-t}$ decreases with p. Its inversion yields (5) and satisfies $\mathbb{P}_p\{p > \overline{r}(K)\} \leq \gamma$. Conditional on $\mathcal{D}$, a union bound across the Cn pairs gives simultaneous coverage at least $1 - \alpha$, without requiring independence across pairs. On that event every eligible candidate meets every tolerance, however the final eligible one is chosen. Averaging over $\mathcal{D}$ gives the unconditional statement. □

At zero observed successes, $\overline{r} = 1 - \gamma^{1/B}$. A coordinate can pass tolerance τ with zero successes exactly when

$$B \geq \left\lceil \frac{\log(\alpha/(Cn))}{\log(1-\tau)} \right\rceil. \tag{6}$$

For $C = 4$, $n = 2$, $\alpha = .05$ and $\tau = .12$, this is 40 independent trials per coordinate, or 320 if all eight coordinates receive the same budget. A positive success count can require more. Reusing one auxiliary file across nominal trials, sampling finite targets without replacement, or learning from final audit outcomes requires another valid design.

5. Scale, demonstration and evidence limits

A width-q, depth-L dense model uses $O(dq + (L-1)q^2 + qn)$ parameters and forward operations for $L \geq 1$. At N rows and E epochs training is $O(EN[dq + (L-1)q^2 + qn])$. This does not reduce the Cn simultaneous confidence limits or the attack trials behind them. Even if a true risk function is Lipschitz in d features, a direct accuracy-ϵ grid can require order $(L_0/\epsilon)^d$ cells for Lipschitz constant L_0. Network capacity alone does not remove this dimensional cost.

The package includes a *synthetic* comparison with six input features and two risk coordinates (Table 1). For each of ten seeds it draws three disjoint toy programmes: 320 counted training rows, 100 rows used only for early stopping and 2,000 rows used only to report the loss. A logistic model and a three-hidden-layer ReLU model train on identical trials until 150 epochs pass without improvement; no best epoch reaches the cap. The logistic model has slightly lower test log loss in all ten replicates, and the mean gap is smaller than the spread across replicates. The table shows that the pipeline runs, not that depth helps.

Table 1: Synthetic SENTRY-R comparison over 10 seeded replicates of the toy law. Each replicate draws 320 training rows, 100 rows used only for early stopping (patience 150 epochs, cap 6000) and 2000 test rows used only for the reported losses. Entries are mean (standard deviation) across replicates; smaller is better. The last row is the paired deep-minus-logistic test log loss (range 0.0004 to 0.0050; deep lower in 0 of 10). These are not results on real release programmes.

Proposal model	Test log loss	Test Brier score	Best epoch (range)
Logistic	0.2556 (0.0043)	0.0727 (0.0017)	404–898
Three-layer ReLU	0.2576 (0.0047)	0.0735 (0.0018)	79–402
Deep minus logistic	0.0020 (0.0014)		

These are properties of a designed toy law, whose test programmes share the training law and feature range, and say nothing about accuracy on confidential or public microdata. A valid assessment must hold out whole real release programmes, compare worst-group calibration and utility at fixed audit budgets, and repeat pooled-vintage attacks. The existing credit-register and public-file results cannot be re-labelled as evidence for the new model.

A shift in the full attack-draw law from $P_{\theta j}$ to $Q_{\theta j}$ with justified total-variation bound $\delta_{\theta j}$ permits $r_{Q_{\theta j}}(\theta) \leq \overline{r}_{\theta j} + \delta_{\theta j}$ (up to one) on the same simultaneous coverage event. A change in an embedding is not itself such a bound. Missing attacks, future auxiliary files and out-of-scope input values must be recorded as gaps, not assigned a reassuring predicted probability.

6. Implications for a release committee

The model is a search instrument. Its model file records the feature schema, training-only scaling, weights, training and validation hashes and held-out loss. A frozen shortlist records its own digest, α, the planned trials, the tolerance owner and the expiry date. The decision record then contains every observed audit count, upper bound, risk coordinate, tolerance, utility, owner and expiry. Only measured development attacks can update the network; only fresh final audits can pass the release gate. This division makes an LLM useful for widening scrutiny without granting its generated answer authority over confidential-data release. Source, tests, sample commands and an editable workflow diagram accompany the longer mathematical manuscript.

References

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